Week 5

Course Level Objectives:

- Recognize, examine, and interpret the basic principles of describing and presenting data. (CLO 2)
- Solve linear regression and correlation problems. (CLO 7)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Compare a quantitative variable across groups using side-by-side boxplots. (CLO 2) **2.** Compute and interpret a difference in means. (CLO 2) **3.** Describe the direction, form, and strength of the relationship in a scatterplot. (CLO 7) **4.** Estimate the correlation r from a scatterplot. (CLO 7) **5.** Use technology to compute the correlation r. (CLO 7) **6.** Explain how outliers, nonlinear patterns, and the absence of causation limit the interpretation of r. (CLO 7)	e-Text: 2.4 Two Variable Relationships Slides: 2.4 Two Variable Relationships	Participation: Box Plots and Outliers – Mon: 1.5×IQR rule review; side-by-side boxplots (tea vs. coffee); Your Turn: Tea vs. Coffee: Descriptive Statistics Participation: Two Variable Relationships – Wed: NBA rebounds vs. free-throw % scatterplot; find r with and without an outlier	Online Homework: 2.4 Two Variable Relationships (due Sun Sep 27, 11:59 PM) Take-Home Quiz 3 (due Mon Oct 5, 11:59 PM) – covers 2.4–2.5 (objectives 3–6) Test 1 (Fri Oct 2 – Sun Oct 4, proctored) – covers Intro, 1.1–2.5 (all objectives) Test 1 Study Guide (all objectives) Practice Test 1 (optional; due Thu Oct 1) (all objectives)	WileyPLUS Blackboard Respondus
7. Interpret the slope of a regression line in context with correct units. (CLO 7) **8.** Interpret the intercept of a regression line in context and evaluate whether it is meaningful. (CLO 7) **9.** Use a regression equation to predict a response value. (CLO 7) **10.** Compute and interpret a residual. (CLO 7) **11.** Locate positive and negative residuals on a scatterplot. (CLO 7) **12.** Use technology to find the least-squares regression line. (CLO 7) **13.** Explain what “least squares” means. (CLO 7) **14.** Recognize when a regression prediction is not appropriate: extrapolation, nonlinearity, and influential outliers. (CLO 7) **15.** Explain why a strong regression relationship does not by itself establish cause and effect. (CLO 7)	e-Text: 2.5 Two Quantitative Variables: Linear Regression Interactive Lecture: Section 2.5 Two Quantitative Variables: Linear Regression (Part 1) Interactive Lecture: Section 2.5 Two Quantitative Variables: Linear Regression (Part 2) Slides: 2.5 Two Quantitative Variables: Linear Regression	Examples: baseball game length and commute-time regression lines (predict, interpret slope, find a residual)	Online Homework: 2.5 Two Quantitative Variables: Linear Regression (due Sun Sep 27, 11:59 PM) Take-Home Quiz 3 (due Mon Oct 5, 11:59 PM) – covers 2.4–2.5 (all objectives) Test 1 (Fri Oct 2 – Sun Oct 4, proctored) – covers Intro, 1.1–2.5 (all objectives) Test 1 Study Guide (all objectives) Practice Test 1 (optional; due Thu Oct 1) (all objectives)	WileyPLUS StatKey Blackboard Respondus